

assignments in

Junior Science

book 4

**Ecology/
Coordination**

**CHRIS GREEF
SUADA BILALI**



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Introduction



Each book in the *Assignments in Junior Science* series provides a selection of worksheets covering one or two major topics. The worksheets are independent, supplying most information necessary to formulate answers. In the answer section at the back of each book, sample answers to most tasks have been given.

Many worksheets have an emphasis on literacy, others concentrate on problem solving, while others utilise science in an experiment or a game format. They are presented in a student-friendly manner and could be used to supplement science lessons for regular class teachers or casual teachers.

Suada Bilali and Chris Greef are both experienced and practising science teachers who had a lot of fun putting together this collection of worksheets.

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Body system jumble



This activity is about the circulatory, digestive, excretory and respiratory systems.

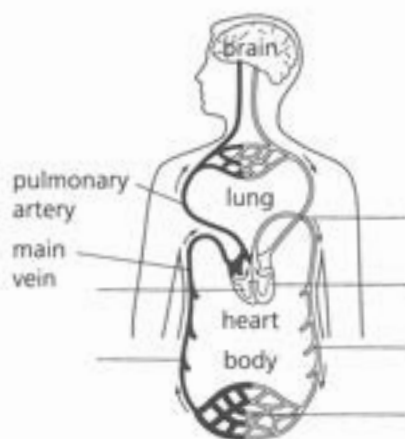
- 1** Complete the table below giving an overview of the systems. However, you cannot use the words *circulate*, *digests*, *excrete* or *respire* in the table.

System name	Main organ	Main function
Circulatory		
Digestive		
Excretory		
Respiratory		

- 2** Below are the parts and functions of the circulatory, digestive, excretory and respiratory systems. Cut out all labels and paste them in the correct positions on the following four diagrams.

Body parts			Functions
alveoli/ air bags	anus	aorta	Breaks down food by physical and chemical means
appendix	artery	bladder	Carries food and oxygen around the body to all cells
bronchioles	capillaries	kidney	Carries wastes away from the cells to lungs and kidney
large intestine	left heart muscle	left lung	Gets rid of unwanted food
liver	nose	oesophagus	Helps regulate water balance and pH = acidity
rectum	right heart ventricle	right kidney	Places oxygen from air into the blood
right lung	salivary gland	small intestine	Releases carbon dioxide and water from blood to air
stomach	tongue	trachea/ windpipe	Removes water, wastes and impurities from blood
ureter	urethra	vein	Stores wastes as urine

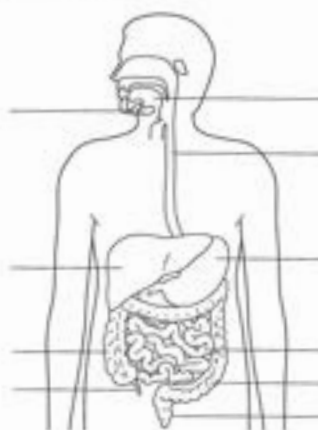
a Circulatory system



Functions

- (i) _____
(ii) _____

b Digestive system



Functions

- (i) _____
(ii) _____



- (i) _____
- (ii) _____
- (iii) _____

d Respiratory system



- (i) _____
- (ii) _____

3 Refer to the information given in the labelled diagrams and to your own knowledge and summarise information about the main human organ systems in the table. The first system has been done for you already.

[illegible]



- ## The main organ systems in the human body

The circulatory system transports food, oxygen, water and wastes through the body. Its main organs are the heart, arteries, veins and capillaries.

The heart is in the middle of the chest and pumps blood in arteries away to all parts of the body. When the blood goes into capillaries, food and oxygen go into the body cells and wastes go from the cells into the blood. Veins take the blood back to the heart. The heart is a muscular pump.

This image shows a full page of blank, lined paper. It features approximately 20 evenly spaced horizontal grey lines across the entire width of the page, typical of standard notebook or legal stationery. The background is a solid off-white color.



In humans the male (produce / produces) sperm, which (fertilise / fertilises) eggs inside the female body.

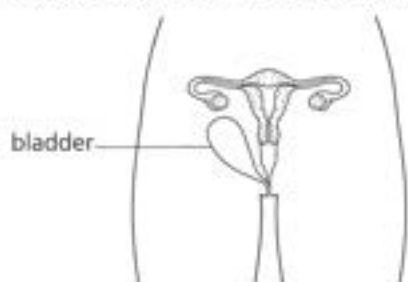
The male reproductive system (consisted / consists) of testicles and several tubules. The two testicles (are / have) nut-sized balls which (are carried / carry) outside the male body in a sac called the scrotum. The scrotum (kept / keeps) the testicles at just below body temperature, so they don't overheat, which would harm the sperm. The testicles (produce / production) sperm continuously after puberty and (store / stored) them in the epididymis pipe. The sperm duct (carries / carry) about 300 million sperm during ejaculation from the epididymis to the beginning of the urethra. The seminal vesicle and Cowper's gland (are / is) small bags which (produce / production) a liquid. This liquid (is / will be) added to the sperm at the beginning of the urethra to (form / formation) semen. The urethra is a pipe which (was / is) inside the penis. It (carriage / carries) the semen from the sperm duct through the erect penis during ejaculation. At other times the urethra (takes / took) the urine from the bladder out of the body.

The female reproductive system (is / was) made up of the ovary, the uterus and several pipes. Each of the two ovaries (are / is) about the size of a walnut. They (are / is) in the abdomen inside the female body and (maturation / mature) one of about 500 000 eggs every month onwards from puberty. The two oviducts (are / were) pipes which (are / is) open near the ovaries and then connect to the uterus. One oviduct (collected / collects) a matured egg from the ovary every month and (allow / allows) it to travel through it. It (is / was) here that the egg (may be / will be) fertilised by sperm. The uterus (have / is) a muscular bag that (receive / receives) the fertilised egg and (let / lets) it embed itself in the swollen wall in order to (grew / grow) into a baby. The vagina (is / was) a muscular tube that (connected / connects) the uterus to the outside. After about nine months inside the uterus the baby (is pushed / pushes) out through the vagina and so (is / was) born. The vagina also receives the erect penis during sexual intercourse and (guidance / guides) the semen into the oviduct for possible fertilisation.

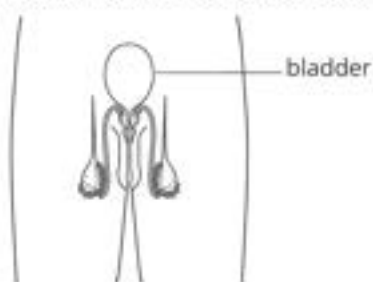
The ovaries (maturation / mature) about 500 eggs during the life of a woman, while the testicles (production / produce) thousands of sperm continuously. Sperm which (are not / are) used during ejaculation (are / were) broken up in the epididymis and are used to (made / make) new sperm again.

1 Read the above text and cross out the inappropriate verbs from the choices in brackets.

2 Refer to the text above and label the diagrams about the male and female reproductive systems.



Female reproductive system



Male reproductive system

3 Complete the table about the parts of the human reproductive systems and their functions.

Name	Function

Name	Function



A dictagloss activity

- 1 A dictagloss is similar to a dictation, but the text to be written is read much faster.
 - The teacher, or another student, will read out Part A (below) to the class while you just listen.
 - Part A is read aloud again and this time you can make some notes on the sheet provided.
 - The text is read a third time while you improve your notes. In groups, make corrections and write the text out in sentences.
 - The same text will then be read for the last time, while you compare and correct your writing.
 - Repeat the dictagloss with Part B.

Note: Teacher instructions to ensure the success of this activity can be found in the answer section.

Part A

The nervous system allows living things to detect what is going on around them. This is called sensitivity. Any change which the nervous system detects is called a stimulus. The nervous system processes the information and then a gland or a muscle reacts. This reaction is called a response.

The nervous system is made up of the central nervous system and the peripheral nervous system. The central nervous system includes the brain and the spinal cord for many involuntary and automatic responses. The peripheral nervous system is a network of nerves that spread throughout the body. They connect the nerve endings in muscles, glands and sense organs to the central nervous system.

Part B

Nerves are made up of special cells called neurones. The bundles of neurones look like the wires in a telephone cable. These nerve cells carry electric impulses or messages at great speed around the body. There are three kinds of neurones: sensory neurones carry messages to the central nervous system; motor neurones carry messages from the brain; and connector neurones carry messages in the brain and spinal cord.

Neurones, or nerve cells, are quite strange looking cells. They have a long thin fibre stretching from a cell body. These fibres are called axons. The axons are covered by a myelin sheath for insulation and greater speed. Then there are short fibres coming off the cell body like skinny fingers. These dendrites pick up messages from other neurones nearby and then pass them along the axon.

Notes for Part A

Text for Part A

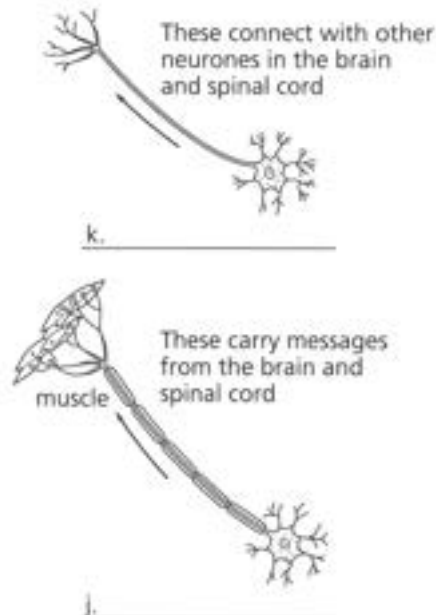
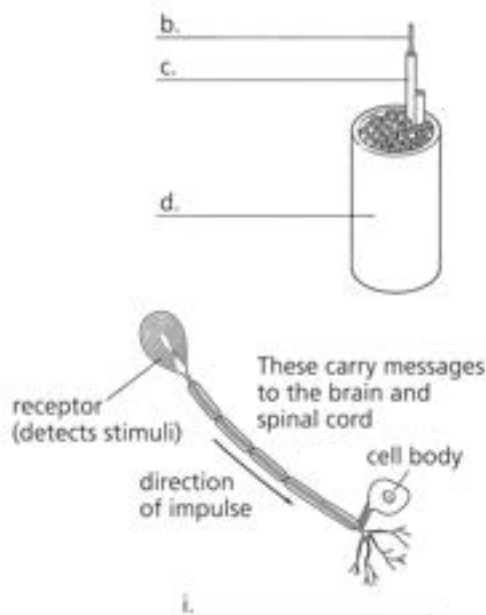
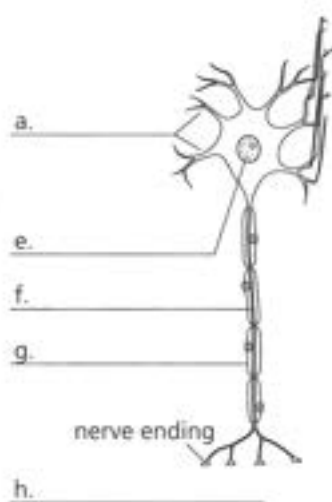


Notes for Part B

Text for Part B

- 2 Use the information from your dictagloss to label each of these diagrams. Choose the labels from the words in the box.

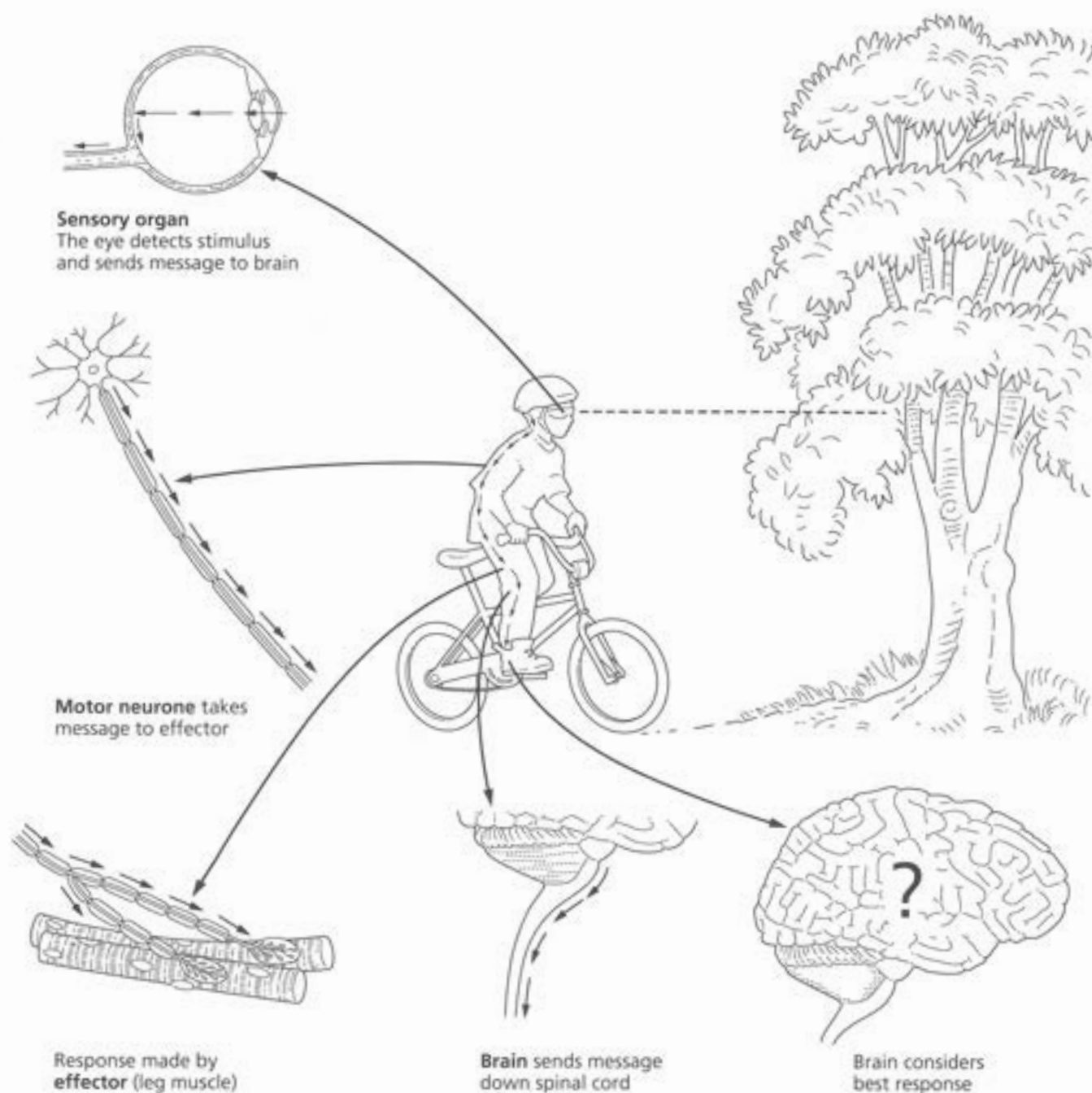
axon dendrites myelin sheath nerve nerve cell neurone
sensory neurone nucleus axon motor neurone connector neurone



Partner A

This is a barrier activity. In order to complete this activity you need a partner. One of you receives part A, the other part B. The aim of the activity is to complete your worksheet by asking your partner questions, without looking at his or her worksheet. Your partner is to complete his or her worksheet by asking you questions, without looking at your sheet. So sit back to back and start asking questions, taking turns.

- a _____ responses to _____ stimuli require _____. This means that the brain is aware of what is going on _____ and decides what action to take. In this _____ of response, _____ is detected by receptors in the _____. _____ send a message along _____ to the brain. _____ decides on a response and sends _____ along motor neurones to an appropriate _____. Glands produce _____ or muscles _____.
- b Ask your partner questions to help you number the diagrams in the correct order.



Voluntary responses

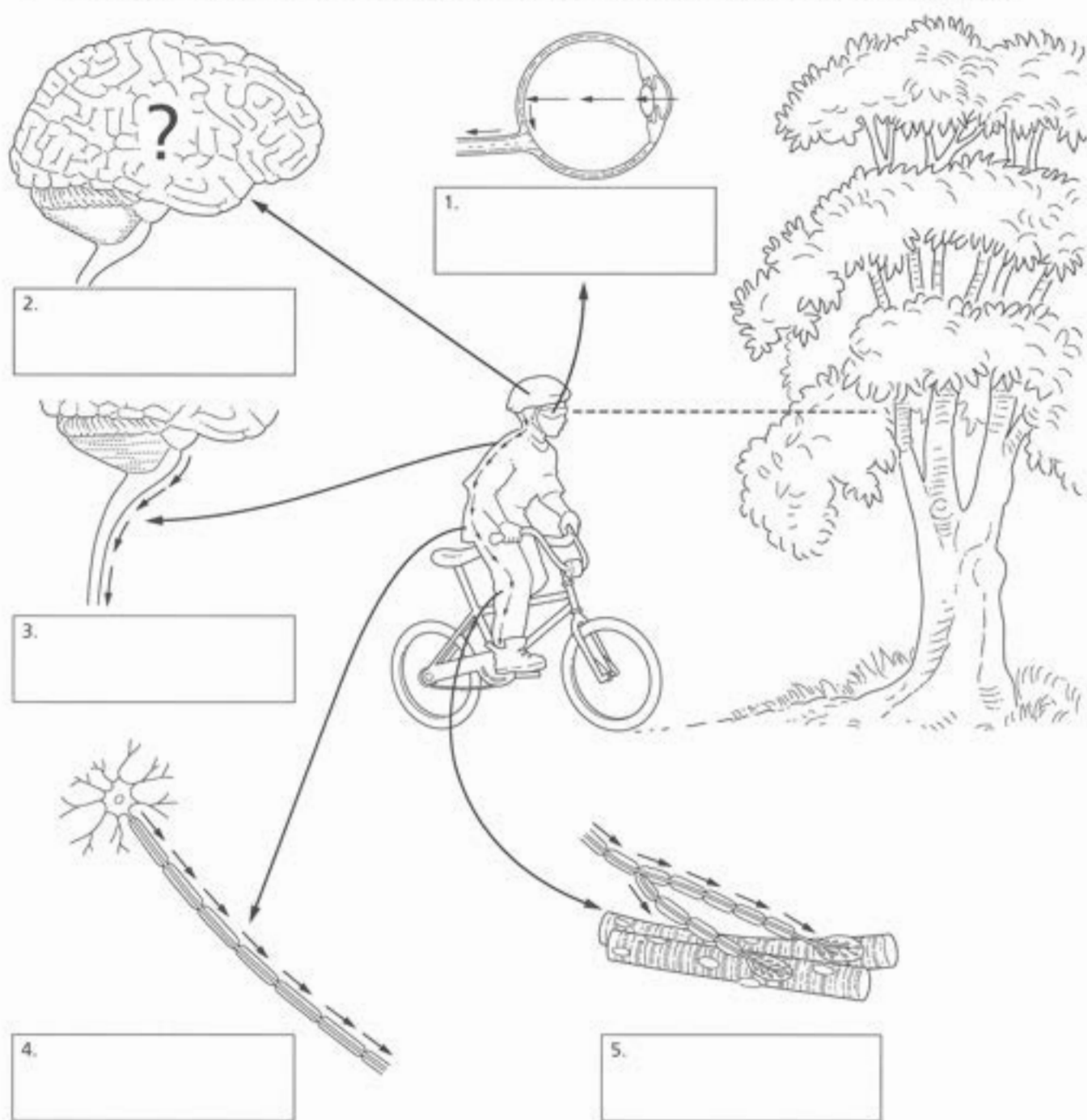
4b

Partner B

This is a barrier activity. In order to complete this activity you need a partner. One of you receives part A, the other part B. The aim of the activity is to complete your worksheet by asking your partner questions, without looking at his or her worksheet. Your partner is to complete his or her worksheet by asking you questions, without looking at your sheet. So sit back to back and start asking questions, taking turns.

a Voluntary _____ to outside _____ require the brain. This means that the brain _____ what is going on around us and _____ what action to take. In this type of _____, the stimulus is detected _____ in the sense organs. The receptors send _____ along sensory neurones _____. The brain decides _____ and sends the command along _____ to an appropriate effector. _____ produce chemicals or _____ contract.

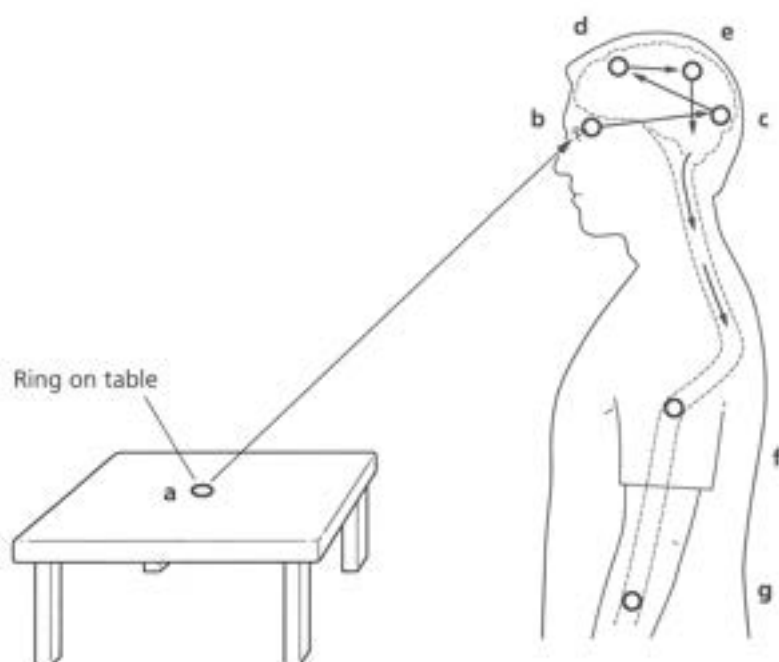
b Your diagrams are in the correct order. Ask your partner questions to help you label each diagram.



How a voluntary response works



- 1** Look at the diagram below, showing the nervous system. Cut out the descriptions on the right and label the diagram.



- A** The think area in the brain decides what to do with the ring.
- B** The sight area decodes the messages and the person sees a ring.
- C** Light reflected off a gold ring stimulates receptors in the retina of the eye.
- D** The brain decides to pick up the ring and sends a message to the spinal cord.
- E** Optic sensory neurones send messages to the sight area of the brain.
- F** Motor neurones take the message to an effector.
- G** The arm muscles contract and the person picks up the gold ring.

- 2** Complete the explanation text by writing the explanation sequence. Use the statements from above as well as the diagram as a basis for your explanation. Join clauses with conjunctions, such as *when*, *because*, *as a result*.

Explanation title	A voluntary response
Summary	In a voluntary response the brain decides on a particular response to an external stimulus.
Sequence	

Involuntary responses



- 1** This information about involuntary responses has been mixed up. Read through the information and arrange the statements in the correct order by numbering them.

a a signal is sent along sensory neurones to the spinal cord.	b are asleep. Reflexes occur the following way: A stimulus
c aware of the movements that we are making	d by the brain. We do not usually become
e effectors. Consequently the reflex action is produced.	f Examples of reflexes include the fast eye blink when a fly
g in a reflex action. Reflexes can happen even while we	h In the spinal cord connector neurones switch in the normal
i Involuntary responses are called reflexes. Involuntary	j is detected by receptors in a sense organ and as a result
k is just about to fly into your eye, and the mouth watering	l or not. They are a very fast kind of
m response and then pass the message to the appropriate	n response, since they do not involve any decision making
o responses happen whether we want them to	p when you see or smell a yummy piece of food.

- 2** Cut out the sentence parts, arrange them neatly into three paragraphs and glue them onto a sheet of paper. You may need to cut some of the sentence fragments in half to make logical paragraphs.

- 3** Briefly write the function of each paragraph.

Looking at language

- 4** Verbs are central to a sentence because they tell us what is going on, what action is taking place. Verbs often show how a participant relates to something else. There are different kinds of verbs:

- Linking verbs connect things with their characteristics. The most common linking verbs are *are* and *have*. Often a linking verb can be replaced with a form of *are* or *have*.
- Action verbs describe the action that is occurring in the clause, for example *run* and *explode*.
- Saying verbs show what somebody or something is saying, for example *shout* and *show*.
- Thinking verbs describe what somebody is thinking or feeling, for example *conclude* and *decide*. Saying and thinking verbs are quite rare in scientific writing.

- a** In your ordered paragraphs above, underline with a straight line any action verbs, underline with a dotted line any linking verbs, circle any (saying) verbs and underline with a wavy line any thinking verbs. Note that verbs can be more than one word long.

- b** State which verb type dominates each paragraph. _____

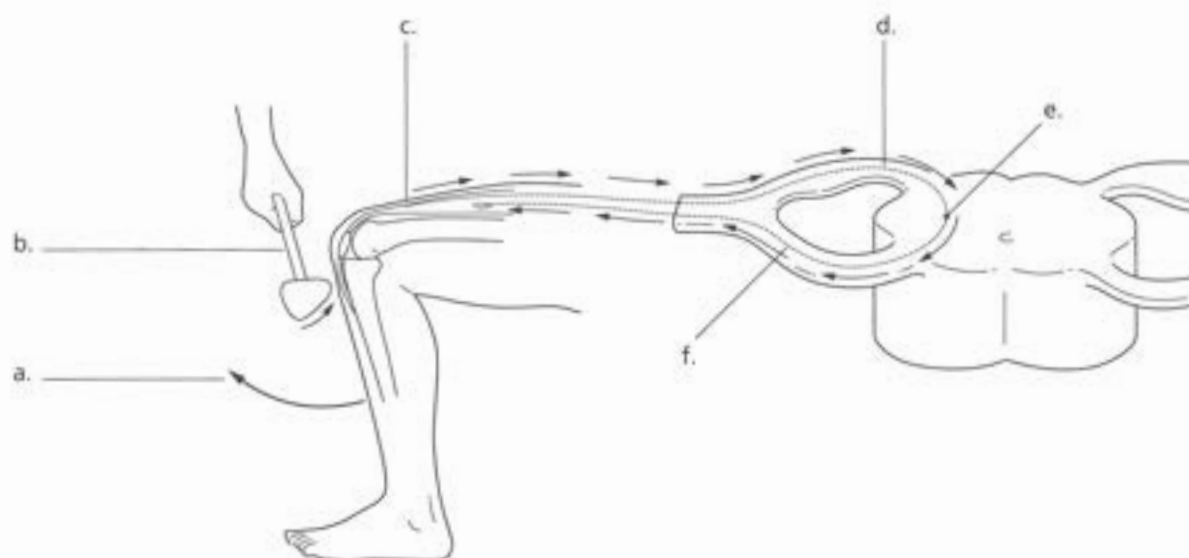
Involuntary responses

6b



5 Cut out each square below and use them to label the diagram about the knee jerk reflex.

A RESPONSE: the knee straightens as leg muscles contract.	B Sensory neurons carry the signal from the RECEPTOR.	C Motor neurons carry the message to the EFFECTOR.	D Stretch RECEPTORS in the leg muscle fire off a signal.	E Connector neurones connect to motor neurones in the spinal cord.	F STIMULUS: a tap stretches the tendon below the knee.
---	--	---	---	---	---



6 Order the events described in the squares above in their correct order, starting with the stimulus.



Information about infectious diseases is presented in text and table format below. However, both sections are incomplete. Use the information given in both sections to complete the text and the table.

Report on infectious diseases

Infectious diseases are caused by different pathogens that invade the body. These pathogens are parasites that live off the host's body and usually cause the host to suffer. Treatments and preventions of infectious diseases depend on the type of organism that the pathogen is. There are different kinds of organisms causing disease: bacteria, viruses, fungi and protozoa.

Bacteria are simple microscopic organisms that are found almost anywhere. They are present in your home, all over your skin and on the food you eat. Bacteria have different shapes such as a rod, a ball, a curve and a cork screw. Some bacteria produce a toxin that poisons and harms your body. Symptoms of a bacterial infection may include coughing, vomiting, fever, diarrhoea and skin rashes. Many bacterial infections are prevented by strict hygiene, especially when handling food or surgical and dental instruments. To treat bacterial diseases antibiotics such as penicillin are administered which help kill the bacteria. Examples of bacterial infections are cholera, impetigo, typhoid, whooping cough and food poisoning.

1 Viruses are much _____ and simpler than bacteria and come in a variety of shapes: some look like _____ or spheres, others have quite complex geometric shapes. Viruses invade the body cells of the host and then take control of the cell. As a result the body cells can no longer perform their normal functions and instead help the virus to reproduce. Symptoms of viral infections range from fevers and _____, to coughing and _____. Viruses are difficult to get rid of, because they are not separate living things. Therefore antibiotics are no use against them. Usually the body needs to _____ so it can fight the disease by itself. Prevention is much better and immunisation is successful against most viral diseases. Examples of viral diseases include _____, flu, _____, cold sores, _____ and chicken pox.

2 Fungi are parasitic organisms which obtain their food from other living or non-living things.

3 Protozoa _____



4

5 Summary table of infectious diseases

Cause of disease	Appearance	Action	Symptoms	Treatment, prevention	Examples
a _____					
b _____	Smaller, _____ than bacteria Rod-shaped, ball-shaped, _____	Take over control of _____ which can't do their job then	_____, Pain in limbs _____, Skin rashes	No treatment Rest _____	Measles, _____, Polio, _____, AIDS, _____
Fungi	Multicellular No roots, stems, leaves	Produce enzymes which digest body cells	Red, scaly, itchy patches on feet, arms, scalp, armpits	Fungal antibiotics, cream or powder	Tinea, Ringworm, Thrush, Athlete's foot
Protozoa	Unicellular animals Larger than bacteria	Live in body cells or blood cells	Diarrhoea Fever Headache Nausea	Anti-malaria drugs (eg quinine)	Malaria, Giardiasis, African sleeping sickness, Chagas disease
Parasitic animals	Tapeworm Mite Flea Tick Liver fluke	Use body as source of food Irritates body	Pain Itchiness Weight loss Diarrhoea Anaemia	Drugs Hygiene Cleanliness in house, clothes	Hydatid disease, Scabies, Flea bites, Blood fluke disease



The table below is a summary of information about non-infectious diseases. Following the table, a report about non-infectious diseases has been started for you. Refer to the table to complete the report, and then answer the questions.

Disease cause	Examples	Prevention, treatment
Inheritance	Cystic fibrosis Down's syndrome Haemophilia (bleeder disease) Sickle cell anaemia Rh syndrome	Maternal screening, eg amniocentesis at 14 weeks into pregnancy Genetic counselling for parents Gene therapy to replace missing or defective gene Decision not to have children if in high risk category
Poor diet	Malnutrition Starvation Obesity Alcoholism Scurvy Arteriosclerosis	Counselling Eating a balanced and adequate diet Drinking alcohol in moderation Educational programs Exercise
Body malfunction	Heart attack Hole in the heart Diabetes Skin cancer (melanoma) Hypertension (high blood pressure)	Eating a proper diet Exercise Medication Replacement of organ Sunscreen, hat
Environment	Lung cancer Asbestosis Deafness Lead poisoning Stress Drug addiction	Quit smoking (various methods & counselling available) Education Changed industrial practices Legal standards and maximum levels Counselling for drug users

Report on non-infectious diseases

Many diseases are non-infectious, which means that your body has not been infected by a pathogen. As a result the disease cannot spread to other people. Non-infectious diseases can be caused by inheritance, poor diet, body malfunction or environmental factors.

A non-infectious disease can be caused by inheritance, where a chromosome or gene is faulty. This can be passed on from an afflicted parent or occur during the reproduction process.

Other non-infectious diseases are caused by a poor diet where not enough nutrients are available or where some nutrients are not eaten at all.



Disease can be caused when body organs or body tissues don't work properly.

Another cause of non-infectious diseases can be the environment we live in. This often relates to chemical and acoustic pollution, as well as stress.

Questions

- 1** Explain how a poor diet can cause both malnutrition and obesity.

- 2** Select one disease from each type of cause shown in the table above, and explain for each how you can prevent developing the disease.



Refer to the labelled diagrams below offering information about hormones and glands in the human male and female body.

Female

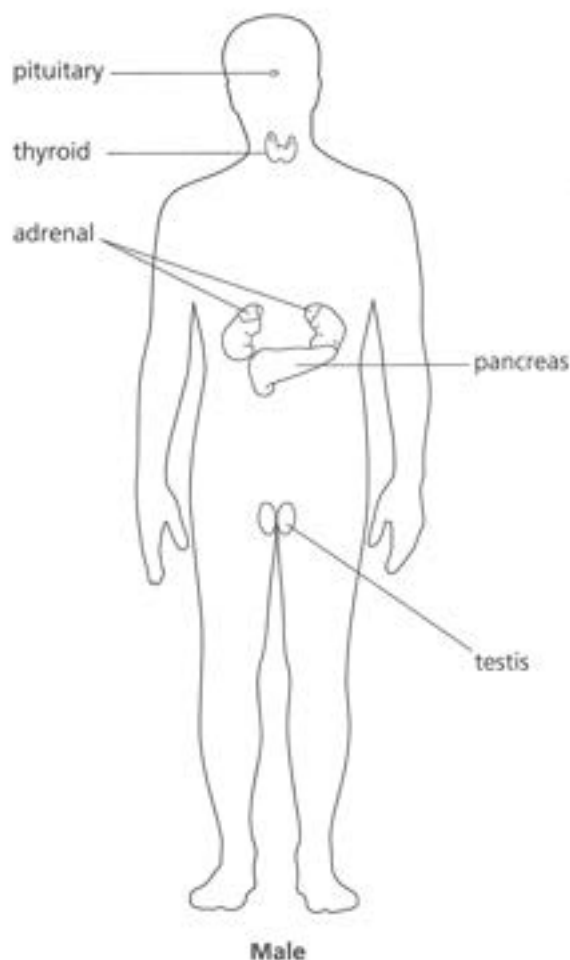
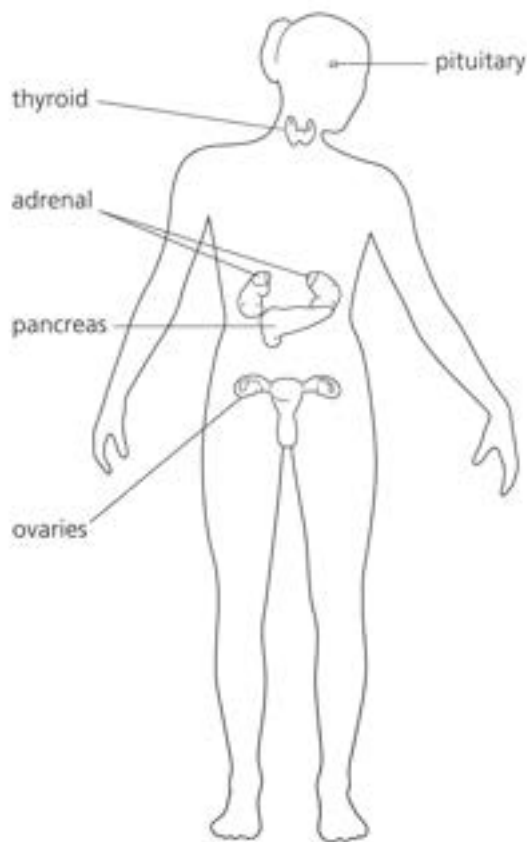
Pancreas Special cells found in the pancreas, called Langerhans, produce the hormone called insulin. **Insulin** lowers the level of glucose in the blood, causing the liver and muscle cells to convert glucose to glycogen.

Pituitary gland is a very small gland, about the size of a pea. It produces human growth hormone which regulates growth in children.

Thyroid gland produces a hormone called thyroxine, which controls energy release in tissues.

Ovaries produce the sex hormones called oestrogen and progesterone which work the menstrual cycle and also develop the female body.

Adrenal gland produces adrenaline, which prepares the body for action by targeting the muscles, heart, liver and blood vessels.



Male

Pituitary gland

- found at base of brain
- master gland
- controls all other glands
- lack of growth hormone could stunt growth permanently

Pancreas

- produces insulin
- not enough or too much insulin results in diabetes

Thyroid gland

- located in neck
- produces thyroxine
- controls metabolism (the rate at which tissues use energy)
- lack of iodine can result in a disease called goiter (enlarged thyroid)

Testis

- produces sex hormone, testosterone
- controls sexual function throughout the male body
- develops male body characteristics

Adrenal gland

- found near top of each kidney
- adrenalin is produced when stressed or excited
- adrenalin causes increase in heart beat, breathing, blood pressure and sugar levels in the blood

[illegible]

2 Refer to your table and to the diagrams on the previous page and write reports on any three glands.

[illegible]



- 1** Below are four paragraphs of text about the two coordination systems in humans. Order the paragraphs by writing numbers in the *Order* column below.

Order	Paragraphs
	Both systems carry information from one place to another. Nerves are very quick. Neurones carry electrical impulses along their axons which then send transmitter chemicals from one axon to another. The impulses travel very quickly along the long axons, but the transmitter chemicals are much slower as they travel across the tiny gap between the dendrites. On the other hand the endocrine system places chemicals called hormones into the blood stream, which then takes them around the whole body at a fairly low speed. Several organs respond to the hormones, while others do not. Those target organs respond fairly slowly and the response can last for a long time, anything from a few minutes to a few months. However, the nervous impulse results in an immediate response which is short lived. The nerves take a message directly to a particular organ or tissue which alone responds.
	Here are some examples of responses involving one system or the other: When the light receptors in your eye relay a close and fast moving insect to the brain via the sensory neurones, the connector neurones in the central nervous system send a message along the motor neurones to the eye lid muscles. As a result of this reflex the lid closes, thus protecting the eye. When something hot or sharp touches you, you very quickly pull back that touched part of the body. Only nerves are involved in this quick and direct response. However, when the thyroid gland in your neck releases its hormones into the blood stream, many parts of your body react over the next few weeks and you grow taller and/or bigger. Another hormone is insulin. When insulin is released from the pancreas into the blood system, the cells around the body absorb glucose sugar from the blood. This glucose is then used for all kinds of cellular processes including respiration.
	In humans the nervous system and the endocrine system work together in order to control the body. Sometimes the nerves bring about responses, other times the hormones. How are they different? How do they work together?
	The fright response involves both systems. This response can be initiated by shock, anger, passion, anxiety or stress. The nervous system detects fright stimuli, which are relayed to the hypothalamus inside the brain. This stimulates the pituitary gland to release hormones into the blood stream. In particular, the adrenal glands above the kidneys react and push the hormone adrenalin into the blood. As a result your pupils dilate, you become pale, your sense organs become highly sensitive, your heart beats faster, glucose is released into the blood, your blood pressure increases and your lungs work more effectively. Your body is now ready for action: either for flight or for fight! Many mammals also have this fright response.

- 2** Each of the four paragraphs above covers one or more topics. In the table below, reveal the structure of the text by linking the main topics and subtopics to the ordered paragraphs.

Paragraph	Main topic	Subtopics

- 3** a State whether the text above is a discussion, an explanation, a procedure, a recount or a report.



b Justify your answer. _____

- 4 Make a summary table showing similarities and differences between the nervous and endocrine systems. Use the *Aspect* column to identify the area you are looking at, then fill in the specific features of the *Nervous system* and the *Endocrine system*.

Aspect	Nervous system	Endocrine system
Purpose		
Speed		
Messengers		
Pathway		
Response organs		
Response activity		
Examples		

- 5 Discuss whether the nervous system or the endocrine system is more important to the human body. Use the scaffold below to help you. Remember to use scientific understanding to your advantage and to include exaggerated and emotional terms.

Title	
Issue	
Arguments for nervous system	
Arguments for endocrine system	
Conclusion	

- 1 Read through the information below and circle the correct word in the brackets.

Ecology (is / are) the study of living things and the way they relate to (there / their) environment or surroundings. The environment (include / includes) things like oxygen, (carbon dioxide / carbondioxide), water, (temparature / temperature), soil, plants, animals and even food.

(Organs / Organisms) can occupy different (enviroments / environments). For example, some organisms (live / lived) on land and others live (on / in) water. No matter (were / where) they live, they all interact (for / with) the living and non-living parts of (their / there) environment.

The living parts (off / of) an environment are called (biotic / bionic) and the non-living parts are called abiotic. These non-living and living parts of an environment make up (a / an) ecosystem. Ecosystems can be (vary / very) large like an ocean or very small like a puddle. Each ecosystem, however, (have / has) a range of living things (which / witch) live (in / on) it.

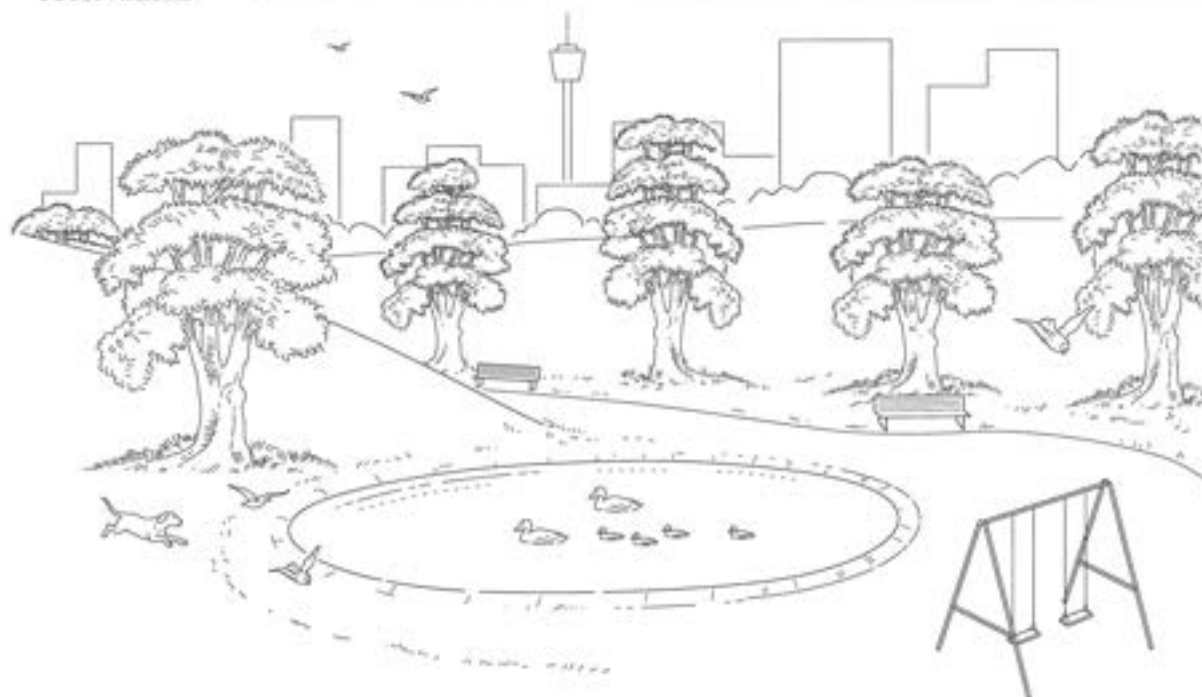
(Within / Without) the ecosystem every living thing has (its/ it's) own special place (to / too) live; this is called its (habitat / habit). The habitat that organisms live (for / in) provides them with all (there / their) needs. A group of organisms (of / off) the same kind, surviving and reproducing amongst each other (in / of) the same habitat, is called a population.

Different groups of organisms that share a habitat make up a (comunity / community). These organisms (need / knead) each other and provide each other (for/ with) shelter and food.

- 2 Complete the table referring to the text above.

- 3 Observe the diagram below. Classify the things in the diagram as biotic or abiotic. Fill in the table provided to record your observations.

Term	Definition
Abiotic	
Biotic	
Community	
Ecology	
Ecosystem	
Environment	
Habitat	
Population	





Table

- 4 A group of students in Ms Smart's class visited the park shown in the diagram. They took some measurements in three different areas in the park. Their results are shown in the table below.

Aspect	Area 1	Area 2	Area 3
Temperature (°C)	18	26	22
Humidity (%)	–	30	50
Light intensity (1 (min) → 10 (max))	3	10	6
Wind speed (km/h)	0	11	7

- a State which area is:

- i) in the pond _____
 ii) under the shade of trees _____
 iii) in the middle of the oval _____

- b Suggest several types of organisms you would expect to see in these three areas.

i) Area 1

ii) Area 2

iii) Area 3

- 5 Describe the specific biotic and abiotic features from one of the three areas.

- 6 List what other abiotic things could be measured in the pond ecosystem and suggest possible measurements.



All in an ant's day – a narrative

There once lived a small colony of ants in a nice comfortable nest under the ground, hidden by the leaf litter of a big gum tree. They were always a busy group going on with their day-to-day business in an orderly manner. Like any other day, this day the ants started by receiving a message, a signal giving them the location of a new stash of food. Many juicy mites had been located high up in their big eucalyptus tree extracting nourishing sap from new leaves. Quickly the ants made their way out of the nest and up along the trunk of the tree.

Along their journey they met several insect borers munching away on some bark. A stick insect curiously moved towards the ants but suddenly stopped. It camouflaged itself on the bark of the tree by staying very still, so it looked like a leafless branch. The surprised ants suddenly understood why the stick insect interrupted its journey when a magpie swooped down and snatched the stick insect for its breakfast. There was not much the ants could do but to continue their journey, wondering whether the stick insect had enjoyed a last meal of fresh leaves.

Continuing their expedition, the ants noticed a young possum enjoying the sweet taste of the tree blossoms. But the possum was rudely interrupted by a cat climbing up towards the branches, hoping to score a possum meal! The possum hurried off, scaring the magpie away from its nest, leaving her young chicks to the fate of the cat or the tree snake that lay absolutely still amongst the branches. 'What an advantage it is being tiny,' thought the ants as they continued to scurry along their path.

It wasn't long till they reached the juicy but clumsy mites. The ants skilfully surrounded the mites and together in their gangs started to carry the mites down the tree. 'Such strength and determination!' thought a flying fox, hanging upside down enjoying the fruits of the tree. It didn't take long for a wagtail bird to notice the colourful mites being hauled down the tree and it flew near the ants to investigate. 'What a great meal!', and with its sharp beak the bird grabbed and swallowed about three mites, along with the ants that had held them. The rest of the ants took one more look at the incoming beak and decided to get out in a hurry – even if that meant losing the mites. Tired and hungry, the surviving ants made their way down the tree towards their nest, only to find it was blocked by some droppings from the flying fox.

The ants retreated into the bark of the tree, where the borers had been previously, to shield themselves from the cold evening. Watching their nest from the safety of the tree and hoping the dung beetles would hurry up and eat the droppings before it got too dark, the ants saw an echidna making its way towards their nest. It all happened so frighteningly fast: the echidna found their nest, pushed its snout into the nest and with its long sticky tongue scooped up every last ant. The ants in the tree could not help but think how lucky they were this time and resolved never to complain again about another flying fox or its droppings! They set off together to start another colony under another tree.

1 Use the narrative to construct a food web with Diagram 1 on the next page. Cut out the animals mentioned in the story and paste them in the correct position on the tree house. Join the animals by arrows according to their eating patterns.

2 Look at your food web and list the:

- a producers _____
- b first order consumers _____
- c second order consumers _____

3 Construct three food chains from the food web.

4 Predict what would happen if all the birds living in the tree were killed by the cat.

Look at my tree house

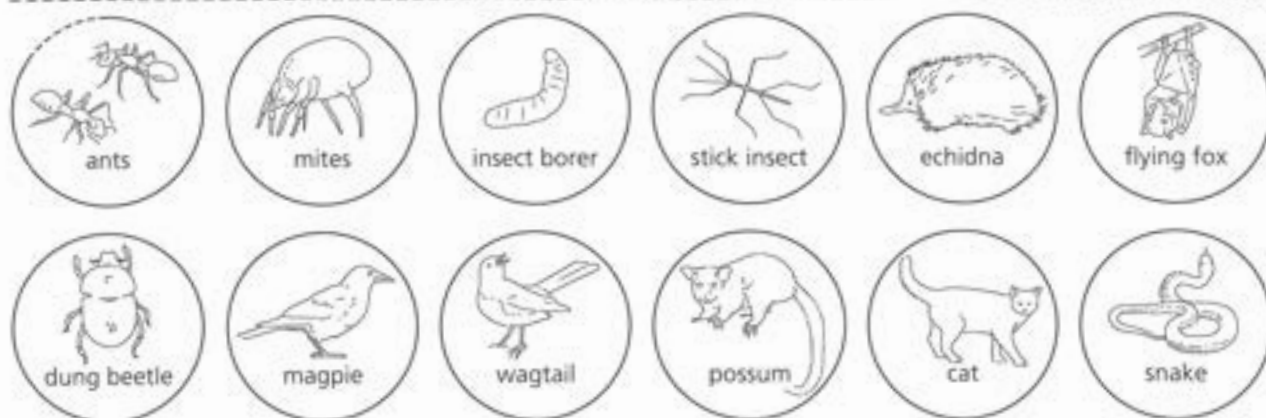
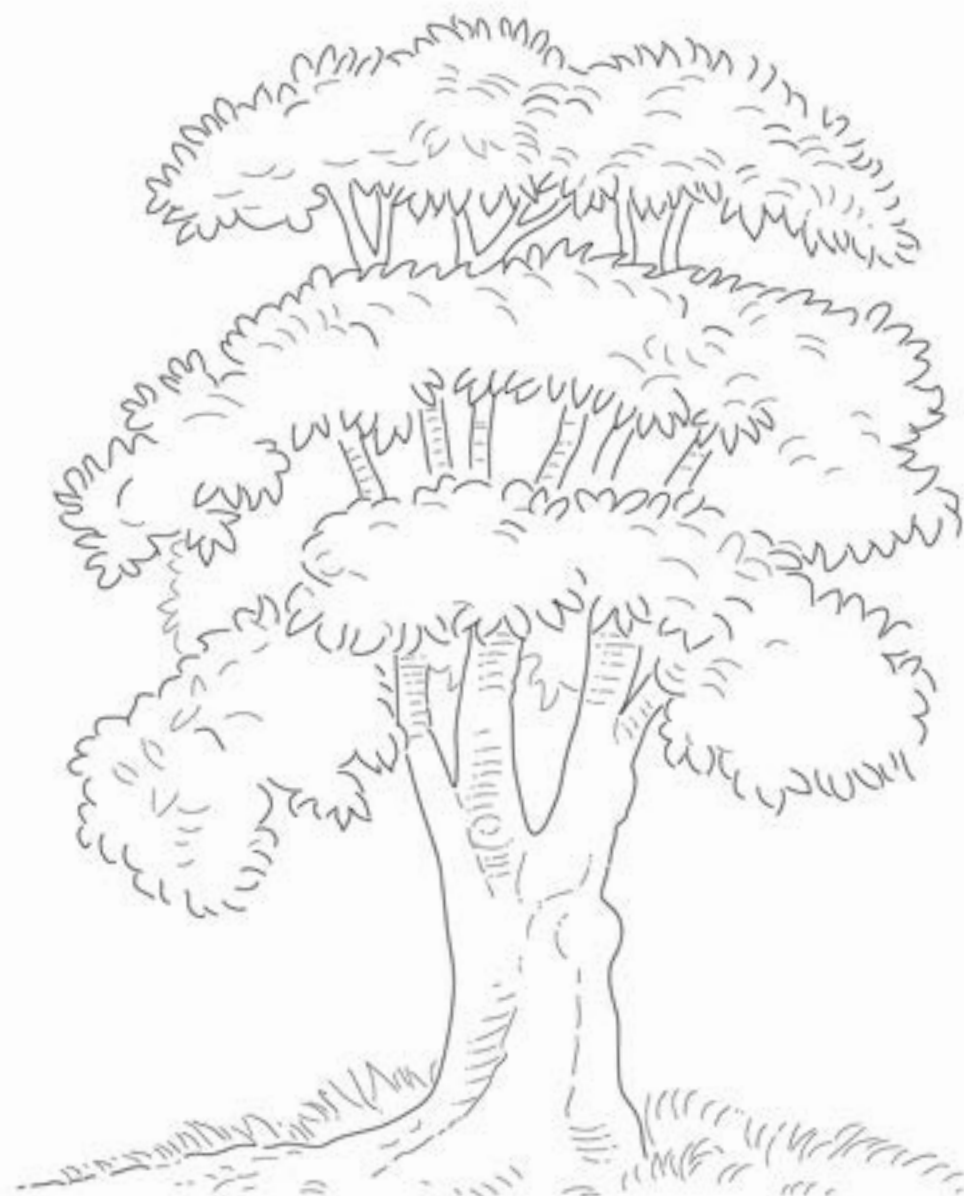
- 5 a Connect these food chains into a food web.

Algae → tadpole → small fish → duck

Pondweed → snail → frog → kookaburra

- b Write your own narrative using your food web.

Diagram 1





1 Read the information below about the life cycle of a nitrogen molecule. Concentrate on the use of pronouns, some of which are underlined and numbered in the text. A pronoun is used instead of a noun (the term *pronoun* means *for a noun*). In the table that follows, list each of the underlined pronouns with their number and write down the noun or noun group it has replaced. Note that sometimes *it* does not refer to a noun, eg *it* is raining (no noun).

As you¹ may be aware, the air in our atmosphere is made up of 70% nitrogen gas. Plants are not capable of absorbing it² from the atmosphere. Carbon, by contrast, can be absorbed as carbon dioxide. Plants obtain nitrogen in a different way, as this story will reveal.

The year is 1750 and Benjamin Franklin is up to his³ creative electrical experiments again. This time he⁴ is flying a kite during an electrical storm. However, this story is not about Benjamin, but about a molecule of nitrogen, called N_2 , that was breathed out of Benjamin's mouth on this big night. We⁵ will explore the nitrogen cycle by taking an imaginary journey with N_2 .

Being a windy night, N_2 was picked up by a rush of wind and carried up over the open field where Benjamin was flying his⁶ kite. It⁷ ended up high in the air, right next to the kite. Suddenly a bright flash of lightning caused N_2 to oxidise. It was broken up and then joined together with two oxygen atoms. To celebrate this unexpected change, its⁸ name became NO_2 . Continuing on its hurly-burly journey NO_2 crashed into some raindrops. Again it was involved in a chemical reaction: it⁹ absorbed water, formed an acid and straight away split into the molecule fraction called nitrate. And again its name changed, this time to NO_3 . Like most raindrops this one too fell to the ground. NO_3 was knocked about in the soil, being dragged deeper and deeper by the rainwater.

Finding itself¹⁰ now surrounded by soil particles and millions of bacteria, NO_3 was quickly and easily absorbed by a plant's root. It was pulled up through a long thin tube, known by the name of xylem, making its¹¹ way eventually into a cell at the tip of a young shoot. NO_3 was moved around in the cell and before it knew it¹², it¹³ had become part of a huge molecule. Being part of a plant molecule was cause for celebration and a new name: Protein.

Protein liked it in the cell: it¹⁴ was contributing constructively to the performance of the cell and was well-liked by the other molecules. But all good things must come to an end, and the young shoot growing under the sun was eaten by a grasshopper, which in turn was eaten by a bird. Protein passed through the bird's digestive system, then its¹⁵ blood system, and was finally used to repair some muscle from a previous injury.

The bird was on its¹⁶ migration towards the Southern Hemisphere but it was scared to death by a sudden flash of lightning and the most enormous thunderclap. Completely dead, it¹⁷ fell to the ground. In the warm morning flies, beetles and other insects congregated around the carcass. Then the buzzing and eating began. Fortunately Protein managed to be overlooked by the insects, but it was not overlooked by one of the millions of microbes. Protein was engulfed by a bacterium, absorbed into its¹⁸ tiny body and used up. When it was finally released into the soil, it had changed again. Its¹⁹ new name was NH_3 . Now it²⁰ belonged to the group of nitrogen compounds called ammonia. Surrounded once again, but by different bacteria, it was taken up and used as a source of energy. In using this energy they²¹ released NH_3 as a nitrate. It²² was nice receiving one of its²³ old names again and NO_3 did have a nice ring to it²⁴. Surrounded by many other nitrates, NO_3 just missed out being absorbed by a plant's root but it could not escape a denitrifying bacterium. It²⁵ changed NO_3 yet again, this time into atmospheric nitrogen. N_2 remembered how a long time ago it had come out of the mouth of a famous person. In trying to make its²⁶ way back into the atmosphere, N_2 found itself²⁷ caught by nitrifying bacteria which lived in the root nodules of a legume. Of course, they²⁸ changed N_2 – they changed it²⁹ into NO_3 .

It³⁰ has been many, many years since our nitrogen molecule N_2 left Benjamin's mouth. He³¹ has been dead for 200 years and is now considered a famous scientist, being studied in many schools around the world. So what has become of our³² friend N_2 ? N_2 has continued to cycle within the biosphere all this time. It³³ has changed many times and adopted the many known names several times, but the atom of nitrogen has remained the same.

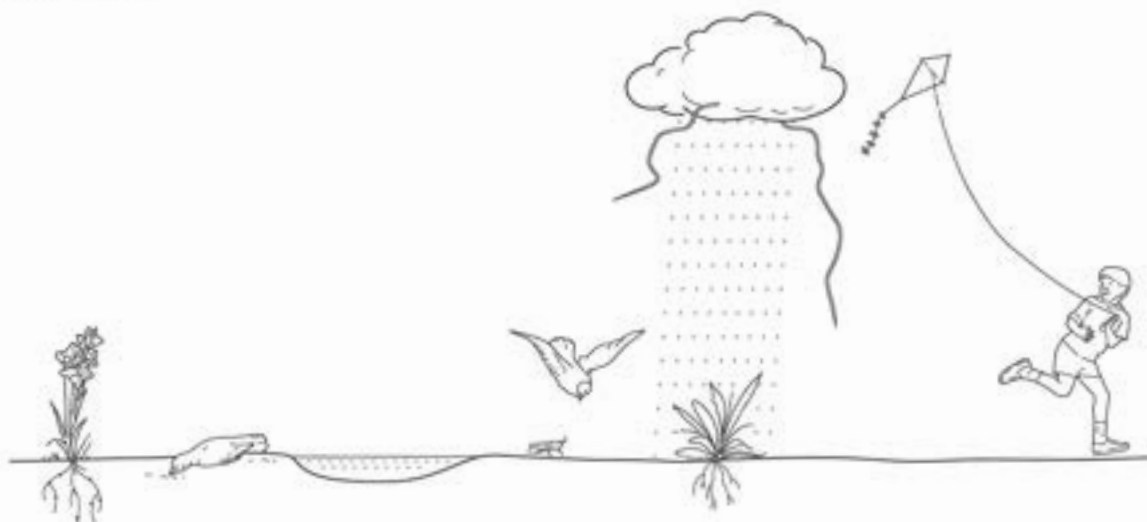


Pronoun	Noun group
you ¹	the reader
it ²	
his ³	
he ⁴	
we ⁵	
his ⁶	

Pronoun	Noun group

Pronoun	Noun group

- 2** Add labels to the diagram of the nitrogen cycle below using the information from the narrative above.



- 3** Change part of the narrative into an explanation. Use the scaffold to guide you. Explain how the atmospheric nitrogen gas (N_2) is changed so plants are able to absorb it.

Title	
Summary	
Sequence	

- 1 Read the text below and use the following words to complete the passage:

green pigment are combined because Chlorophyll most plants
the sun they This energy

All energy needed for life to exist on Earth comes from _____. _____ is trapped by green plants in a process called photosynthesis. During photosynthesis carbon dioxide and water _____ to make glucose (a kind of sugar) and oxygen. For plants to photosynthesise _____ also need a _____ called chlorophyll. _____ is responsible for absorbing light in all producers. This is the reason why _____ are green. Plants are also called autotrophs _____ they make their own food.

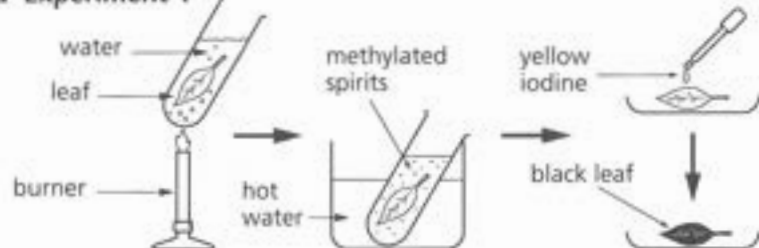
- 2 Define the term 'producer'. _____

- 3 Complete the word equation for photosynthesis below.

a _____ + water $\xrightarrow{\text{b _____}}$ sugar + d _____
c _____

- 4 A science class performed the following experiments to test various aspects of the word equation. Carefully read through each procedure and write what you think the aim and conclusion would be.

a Experiment 1



- A plant leaf was boiled and placed in a test tube of methylated spirits, which was then placed in very hot water.
- When the chlorophyll was removed, the leaf was rinsed and placed on a petri dish.
- Iodine solution was added to the leaf. It turned from yellow to blue/black. Starch was present.

Aim:

Conclusion:

b Experiment 2



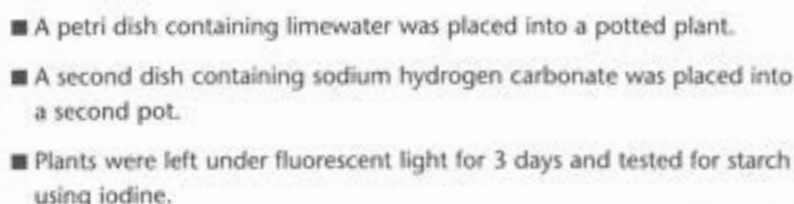
- Using aluminium foil, the top half of individual leaves on a plant were covered.
- The plant was then placed under fluorescent light for 3 days.
- A leaf was then removed and tested for starch using the same steps as in experiment 1.

Aim:

Conclusion:



(Note: Limewater absorbs CO_2 out of air; sodium hydrogen carbonate releases CO_2 into the air.)



Conclusion:

- A plant was exposed to fluorescent light for 4 days.
- The resulting gas was collected in a test tube, which was then corked and tested with a glowing splint.
- Splint glowed brighter.

Conclusion:

- 5** Write up one of the experiments above as an experimental record.

[illegible]

The respiration process – an explanation

Photosynthesis catches energy and respiration releases it.

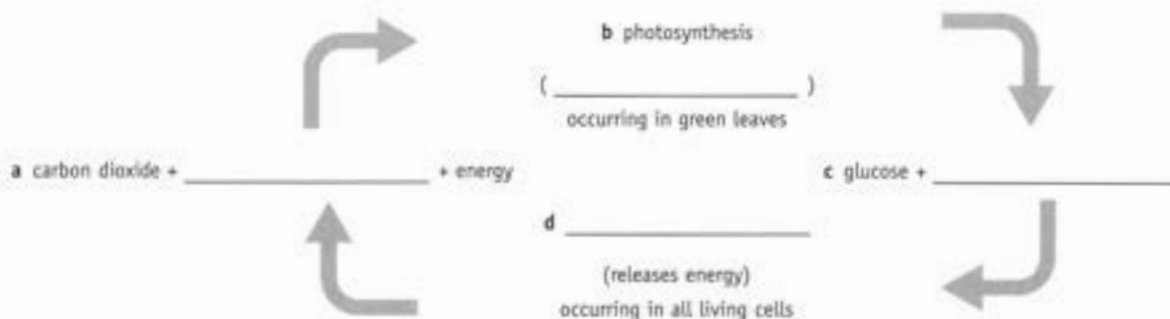
You may have learnt that plants trap energy from the sun and use it to convert low-energy carbon dioxide and water to high-energy glucose and low-energy oxygen. This process is called photosynthesis. Plants store their glucose as starch, a complex carbohydrate molecule. Animals obtain their food by consuming plants or other animals, so they are called consumers. Inside the digestive system of animals, the complex starch is broken down again into a simpler form, glucose. The glucose travels to the cells and is further broken down in the process of respiration. During respiration energy is released and made available to the cells. The cells use the energy for many chemical reactions, as well as for heat.

The sun's energy, which is trapped by the plants during photosynthesis, is released during respiration. In other words, respiration is an energy-releasing process. It occurs in all living things, in their cells. During respiration oxygen is used up as the glucose is broken down in many controlled reactions. Carbon dioxide and water are produced as the waste products. These become the raw materials needed for photosynthesis.

There is a cycle involving photosynthesis and respiration. In this way we do not run out of food, oxygen and carbon dioxide.

- 1 Write the word equation for respiration.

- 2 Complete the diagram by filling in the blanks using the following labels: absorbs energy, oxygen, respiration, water.



- 3 Complete the summary table about respiration and photosynthesis.

Question	Photosynthesis	Respiration
What does the term mean?		
Which organisms perform the process?		
Is energy absorbed or released?		
What are the starting materials?		
What are the end products?		
Why is the process important?		

- 4 Write a paragraph about each process using the information from your table and from the text above.



- 5** A group of students designed an experiment to demonstrate respiration. Read their recount about their experiment.

Our snail experiment – a recount

Our teacher provided us with the following items: a large 500 mL beaker, a large test tube, 3 small snails, some fine wire mesh, 2 small heavy cubes about 2 cm long, and 250 mL of limewater.

To set up our experiment we placed the three snails into the test tube and secured them with the mesh, giving them a little room to move but not allowing them to fall out when we inverted the test tube into the limewater. The open end of the test tube was balanced carefully on the blocks, making sure the liquid could come into the test tube. Initially a small amount of water, about 5 mm, entered the test tube. We marked the test tube using a permanent marker. We let the experiment stand for a day. The following day we noticed the water level in the test tube had risen by 2 cm and the limewater, especially in the test tube, had turned milky. We were all glad to see that the snails were still alive. We dismantled the experiment and let them go.

- a** Draw a labelled scientific diagram of the experiment.

- b** Explain why the limewater turned milky.

- c** Explain why the limewater moved up in the test tube.

- d** Change the recount into an experimental record. (You do not need to draw the diagram again.)



Memory challenge game

Cut out all the cards. Spread them out, face down, on the table and challenge your partner to a memory game. Turn over two cards. When you uncover two matching items (ie a term and its definition), the pair is removed, counted as one point and you have another turn. If the two cards are not a pair, they are turned over again and it is your partner's turn. The winner has the most points. Good luck!

Organism	Food web	Decomposers	Green pigment found in plants.	Feature that makes an organism suitable to its environment.
Photosynthesis	Chlorophyll	Community	Place where each living thing lives.	Links showing what is eaten by what in a community.
Respiration	Population	Adaptation	Process where sugar is broken down for energy. CO_2 and H_2O are waste products.	A number of living things of the same type.
Consumer	Biotic	Starch	A living thing.	Group of living things sharing an environment.
Producer	Abiotic	Complex sugar stored in plants leaves.	Study of the way living things relate to their environment.	Living parts of an environment.
Ecology	Environment	A consumer that eats only meat.	Process where light energy is made into chemical energy.	Non-living parts of an environment.
Ecosystem	Herbivore	Living and non-living parts of an environment.	The surroundings of a living thing.	Living thing that makes its own food.
Habitat	Carnivore	An organism that feeds on others.	A consumer that eats only plants.	Organisms that absorb their food from bodies or wastes of producers and consumers.

Food chains show how energy moves through a community by looking at what is eaten by what.

snake 	small fish 	mouse 	grass seeds 
kookaburra 	cat 	grasshopper 	hummingbird 
carrots 	large fish 	algae 	tadpoles 
flowers with nectar 	rabbit 	fox 	grass 

- 1** Cut out each square containing an organism above and arrange them into four separate food chains. Don't forget to include the arrow ($\rightarrow$ = energy goes to, is eaten by). Paste your food chains into your exercise book, labelling each organism as either a producer, first order consumer, second order consumer and so on.

Looking at language

Text can be written in active voice or passive voice.

In active voice the person or thing that is doing something is placed at the beginning of the clause, because that is the main subject of the clause.

For example: *The cat ate the bird.*

In passive voice the person or thing that something is happening to is at the beginning of the clause, because this is what the clause is about. And whoever is doing the action is placed towards the end of the clause, if it is stated at all.

For example: *The bird was eaten by the cat. The bird was eaten.*

- 2** Use your food chains from question 1 to complete the table below by writing clauses in active voice and passive voice. One pair of clauses has been done for you.

Active voice	Passive voice
The rabbit eats the grass.	The grass is eaten by the rabbit.
The snake	The mouse
The small fish	
The mouse	
The tadpoles	
The fox	
The grasshopper	
The large fish	



- 3** Below is a table showing different organisms found in Ms Smart's back yard. She watched them all day and recorded her observations in the table below.

Organism	What it feeds on
Grass	Makes its own food (producer)
Mangrove tree ■ leaves ■ fruit ■ flowers ■ branches	(producer)
Bottle brush ■ leaves ■ flowers ■ branches	(producer)
Bees	Flowers and fruits (herbivore)
Dragonflies	Bees and ladybirds (carnivore)
Aphids	Sugars from leaf veins of mango (herbivore)
Stick insect	Leaves from bottlebrush (_____)
Insect borer	Mango branches (_____)
Ladybirds	Aphids (_____)
Wagtail	Insect borers and stick insects (_____)
Honeyeater	Flowers and fruits (_____)
Kookaburra	Stick insects and dragonflies (_____)
Cat	The birds wagtail and honeyeater (_____)
Flying fox	Mango fruits (_____)
Possum	Mango fruit and bottlebrush flowers (_____)

- a Complete Ms Smart's table; she forgot to add the labels *carnivore* and *herbivore*.
b Use Ms Smart's information to construct a food web in your exercise book.

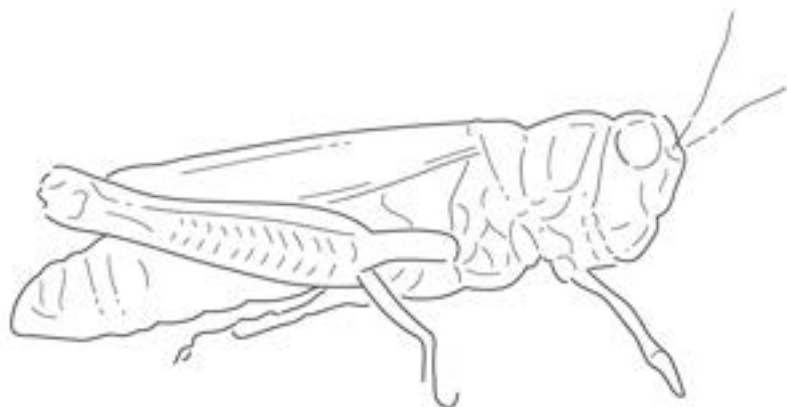
- 4** Ms Smart accidentally sprayed her mango tree and bottlebrush tree with insecticide. As a result all the aphids and insect borers were killed. Explain what could happen to the food web.

- 5** a What would happen if Ms Smart's insect population triples? Explain what could happen to the food web straight away.

- b If the insect population had tripled, explain what would happen over the following weeks and months.



All living things are suited to their living conditions. The features which enable the organisms to live in specific conditions are called *adaptations*. If an organism is not adapted to its environment, it will die there. Adaptations can be behavioural, physiological and physical (= structural). Below is a diagram of a grasshopper and some of its adaptations.



Behavioural adaptations

How does it behave? A grasshopper chooses to quiver in order to mimic the movement of leaves.

Physical adaptations

What is it like? The colour of the grasshopper camouflages it from its predators and its strong hind legs allow it to jump great distances.

Physiological adaptations

How is the body reacting? The grasshopper lays many eggs and is capable of digesting large amounts of plant leaves and shoots.

- 1** Read the information below about the cane toad. Classify its adaptations as behavioural, physiological or physical.

The toad was introduced to Australia to get rid of a pest. It was well adapted to the harsh Australian climate. The toad has rough, brown, dull coloured skin giving it a perfect camouflage against predators. The toad's skin is also very tough with poisonous glands. This deters many predators. Toads also have the capability of puffing up their bodies to make them look larger and more frightening. This scares many predators away. Female cane toads lay large numbers of eggs. They can lay them in fresh or salty water. Toads have been seen to eat a range of insects, and even canned pet food.

- a** Behavioural adaptations: _____

- b** Physiological adaptations: _____

- c** Physical adaptations: _____

- 2** Humans also have many adaptations which make it possible for them to live in many different habitats. For the habitats described below, list in the following table several human adaptations. For each adaptation state whether it is behavioural, physiological or structural and explain the purpose of that adaptation.

- a** a person living in snow country
b a person on the beach in summer
c a person working outdoors in central Australia



Situation	Adaptation	Category	Purpose

Looking at language

Cause and effect conjunctions include such words as *because, since, therefore, so, as a result* and *consequently*. These conjunctions can be used to join short sentences into more complex sentences.

- 3** Complete the table below using cause and effect conjunctions. After constructing each sentence classify each adaptation as behavioural, physiological or physical.

Short sentences	Long sentence	Type of adaptation
a Elephants have large ears. They can cool off quickly.	Elephants have large ears, as a result they can cool off quickly.	physical
b Kangaroos lick their paws. Their spit evaporates. Their paws become cooler.		
c Lizards move to the shade on hot days. They stay cool.		
d Eagles have sharp, curved beaks. They can tear meat.		
e Rabbits rest in their burrows. They stay cool.		
f Dolphins have a streamlined body. They can swim fast.		
g Birds have light bones and wings. They can fly.		
h Kookaburras have a tough beak. They can eat snakes.		
i The cactus has spines for leaves. Spines have no pores. Water is conserved.		
j Dogs shiver on cold mornings. They get warm.		
k Koalas sleep in the middle of the day. They don't get too hot.		
l Frogs have slimy moist skin. They can breathe through their skin.		



The marsupial mole – a report

The Aboriginal name for the marsupial mole varies according to the locality in which it is found. In central western Australia it is referred to as 'the blind sand burrower', *Arra-jarra-ja*, or *Kakarratul* for the northern species and *Itjaritjari* for the southern species. Elsewhere in the southern Northern Territory, it is known as *Oorquamata*.

The marsupial mole looks like a mole or a small solid rat. It has a pouch like other marsupials, measures about 10 – 20 cm and weighs 40 – 60 g. It has soft, fine fur of a cream-yellow colour. It has no external sign of eyes, although there are eye spots hidden under the skin. There is also no sign of external ears, but there are distinct ear openings beneath the fur. The nose is a small shield of horny skin used for pushing the sand away to make a tunnel. The nostrils are small slits. The short tail, only about 2 cm long, is encircled with rings and terminates in a tough lump. The limbs are short and powerful, each foot having five digits. The third and fourth fingers are large and scoop-like, with sharp claws for digging. The pouch opens toward the rear so that sand is not pushed into it.

The marsupial mole is believed not to drink, although it must eat food every day. It seems to reproduce during early summer, but no details are known about its breeding.

- 1 a** Read the report on the marsupial mole and sketch what you think it looks like.

- b** Suggest why scientists do not know about the reproduction of the marsupial mole.

- c** Predict what a marsupial mole would eat, and how it would find this food.

- d** Explain how scientists might have learnt that the marsupial mole does not drink.

Design an animal



- 2** Imagine this environment: It is deep in the ocean, there is only faint reddish light. The floor is covered with large jagged boulders rising up from the sandy ground like mountains.

- a** Design a plant that might grow there. Sketch the plant and explain the purpose of some of its adaptations.

- b** Design an animal that might live in the same environment. Sketch it and explain the purpose of some of its adaptations.

- c** You are a scientist asked to explore this ocean environment. Sketch yourself fitted out and explain the purpose of your clothing and equipment.



- 1** Read the text below about an unusual small bush. In groups, consider each of the cases that follow and discuss possible immediate effects and also likely long-term effects.

A small bush grows deep within a mountainous region. Very little is known about it. Its flowers attract tiny possums which feed on the nectar and fertilise the eggs at the same time. Although possums are nocturnal, eagles manage to catch them for their prey. The fertilised flowers develop into small berries which are part of the diet of a small, shy bird. Although the leaves of the bush are quite tough, new shoots are often eaten by wallabies. Local Aboriginal people sometimes chew young leaves as a treatment for relieving pain.

Case 1: Recently a bush fire raged in the mountains but left the valley of the small bushes untouched. In fact the valley had become refuge for a number of native mammals. There are now fairly large numbers of wallabies, foxes and possums.

Case 2: Birds from another valley have discovered the berries and come often to feed on them.

Case 3: Farmers have built a secure fence around their properties to stop feral dogs from coming out of the bush onto their grazing land.

Case 4: A tree logging company is building a roadway along the valley floor to access large gum trees there and also further away. The trees have many hollows which have become home to many native birds and possums.

Case 5: A thorny bush with sweet berries starts to grow along the edges of the roadway. The berries are eaten by the shy small bird in preference to the berries of the small bush.

Case 6: It's not long before holiday makers discover the valley. Their noise scares the wallabies away and the small shy bird has also withdrawn from the valley. Because the flowers of the small bush are quite pretty and the berries are very sweet, many campers eat the berries and break off little branches with flowers. They also break off branches because they produce a good fire and burn for a long time.

Case 7: The eagles rarely frequent the valley any more because of the many visitors.

Case 8: A pharmaceutical company has discovered the healing properties of the young leaves of the small bush. The company has sent in several workers to harvest several kilograms of young leaves. The company has also applied for a patent on the soothing chemical in the plant. This would require the Aboriginal people to pay royalties every time they chewed the leaves.

Case 9: The abundance of the small bush has become frighteningly low. Rangers have decided to protect the flora and fauna of the valley. Campers have to apply for a permit and entry is only through a narrow pedestrian gate, resulting in fewer visitors. Trucks and cars are no longer permitted at all. Contractors have begun to get rid of the thorny bushes near the roadway.

- 2** Draw a food web below involving as many organisms as possible from the texts above.

Down the slime and up the branches 21a



Environmental issues

- 1** Most countries in the world have signed the *Kyoto Protocol* — Australia and the USA have not. The Protocol commits the country to reduce greenhouse gas emission.

a Suggest ways in which Australians are producing greenhouse gases.

b Suggest ways in which Australians are taking carbon dioxide from the air.

c Justify why Australia *should* sign the Kyoto Protocol.

d Justify why Australia *shouldn't* sign the Kyoto Protocol.

- 2** *Down the slime and up the branches* is a 'snakes and ladders' games based on being friendly or unfriendly to the environment. The game can be played in pairs or groups of up to four students.

Cut out a token for each player and the hexagonal die (or use a proper die if you have one). Colour the token with your identifying colour or pattern.

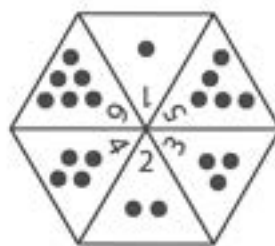
Rules for 'Down the slime and up the branches'

- Place all the players' tokens on START.
- Each person takes turns to roll the die and moves the number of spaces the die displays (flip the hexagonal die in the air and the number pointing to you is the die number).
- If you land on a branch, read the text and climb up the branch as a reward for doing something good in the environment.
- If you land on some slime, read the text and follow the slime down to the very end as a punishment for the bad things that are happening to our environment.
- If you land on a text square, read it and follow its instruction.
- The first person to reach FINISH is the winner provided he or she is able to tell the other players five things we should be doing and five things we should not be doing to our environment within ten seconds.

TOKENS



DIE



Down the slime and up the branches 21b

55	Eutrophication, devastating our waterways	57	In an urban ecosystem, little recycling happens. Move back 3 spaces	59	Increased carbon dioxide levels in the atmosphere due to burning of fossil fuels	61	Too many sheep compacting our soil. Miss 2 turns	62	Washing your car on the lawn. Well done. Move forward 4 spaces	63	Finish
37	Buying things with little wrapping	51	You have a solar heater on your roof. Have a free turn	50	Air pollution caused by heavy emissions from power stations, like in Yallourn Victoria	48	Cutting down trees. Lose a turn	44	Planting trees	43	Recycling old Christmas cards, with 'Planet Arc'. Have a free spin
38	Buying detergents free from phosphates. Have a free spin or move forward 2 spaces	39	Making recycled paper. Have a free spin or move 2 spaces	32	Walking to the corner shop instead of driving	42	You have started a compost heap at home	29	Non-biodegradable rubbish floating on our waterways	25	One cause of global warming is the destruction of the ozone layer
36	Species like the quoll and numbat are becoming endangered due to destruction of their habitat. Miss a turn	21	Sweeping your path instead of hosing it down. Well done. Move forward 2 spaces	14	Oil spills in the harbour	31	Left your lights on when out for the weekend. Miss a turn	12	Buying things made with recycled paper	11	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
18	Clearing of land causes loss of valuable trees	15	Saving water at home. Move forward 4 spaces	4	Recycling cans and paper around your home	24	Walking to the corner shop instead of driving	7	Buying things made with recycled paper	10	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
17	Putting your rubbish in bins	16	Clearing of land causes loss of valuable trees	15	Saving water at home. Move forward 4 spaces	14	Left your lights on when out for the weekend. Miss a turn	13	Buying things made with recycled paper	12	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
16	Clearing of land causes loss of valuable trees	15	Saving water at home. Move forward 4 spaces	14	Left your lights on when out for the weekend. Miss a turn	13	Walking to the corner shop instead of driving	12	Buying things made with recycled paper	11	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
15	Clearing of land causes loss of valuable trees	14	Saving water at home. Move forward 4 spaces	13	Left your lights on when out for the weekend. Miss a turn	12	Walking to the corner shop instead of driving	11	Buying things made with recycled paper	10	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
14	Clearing of land causes loss of valuable trees	13	Saving water at home. Move forward 4 spaces	12	Left your lights on when out for the weekend. Miss a turn	11	Walking to the corner shop instead of driving	10	Buying things made with recycled paper	9	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
13	Clearing of land causes loss of valuable trees	12	Saving water at home. Move forward 4 spaces	11	Left your lights on when out for the weekend. Miss a turn	10	Walking to the corner shop instead of driving	9	Buying things made with recycled paper	8	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
12	Clearing of land causes loss of valuable trees	11	Saving water at home. Move forward 4 spaces	10	Left your lights on when out for the weekend. Miss a turn	9	Walking to the corner shop instead of driving	8	Buying things made with recycled paper	7	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
11	Clearing of land causes loss of valuable trees	10	Saving water at home. Move forward 4 spaces	9	Left your lights on when out for the weekend. Miss a turn	8	Walking to the corner shop instead of driving	7	Buying things made with recycled paper	6	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
10	Clearing of land causes loss of valuable trees	9	Saving water at home. Move forward 4 spaces	8	Left your lights on when out for the weekend. Miss a turn	7	Walking to the corner shop instead of driving	6	Buying things made with recycled paper	5	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
9	Clearing of land causes loss of valuable trees	8	Saving water at home. Move forward 4 spaces	7	Left your lights on when out for the weekend. Miss a turn	6	Walking to the corner shop instead of driving	5	Buying things made with recycled paper	4	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
8	Clearing of land causes loss of valuable trees	7	Saving water at home. Move forward 4 spaces	6	Left your lights on when out for the weekend. Miss a turn	5	Walking to the corner shop instead of driving	4	Buying things made with recycled paper	3	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
7	Clearing of land causes loss of valuable trees	6	Saving water at home. Move forward 4 spaces	5	Left your lights on when out for the weekend. Miss a turn	4	Walking to the corner shop instead of driving	3	Buying things made with recycled paper	2	Air pollution caused by car exhausts. Miss a turn or move back 5 spaces
6	Clearing of land causes loss of valuable trees	5	Saving water at home. Move forward 4 spaces	4	Left your lights on when out for the weekend. Miss a turn	3	Walking to the corner shop instead of driving	2	Buying things made with recycled paper	1	Start



Coordination

1 Body system jumble

System name	Main organ(s)	Main function(s)
Circulatory	heart, arteries, veins	transport nutrients, waste and water through body
Digestive	stomach, intestine, liver	break down food and place it into blood
Excretory	kidneys, bladder	clean blood and produce urine
Respiratory	lungs, mouth, nose	put oxygen into blood and carbon dioxide and water into air

2 The human reproductive system

1 produces, fertilise; consists, are, are carried, keeps, produce, store, carries, are, produce, is, form, is, carries, takes; is, is, are, mature, are, are, collects, allows, is, may be, is, receives, lets, grow, is, connects, is pushed, is, guides; mature, produce, are not, are, make

Name	Function	Name	Function
testicles	produce sperm	ovary	mature eggs
scrotum	keeps testicles below body temperature	oviducts	collect mature egg, site for fertilisation
epididymis	stores sperm	uterus	receives egg, grows baby
sperm duct	carries sperm to urethra	vagina	gives birth to baby, receives penis, guides semen to oviduct
seminal vesicle	forms liquid to make semen	urethra	carries semen to outside, carries urine
Cowper's gland	forms liquid to make semen		
penis	puts semen in vagina		

3 The nervous system

Teacher instructions:

Introduce the topic and the activity. Write on the board the special terms involved (they will be explained in the text but students may be unsure of spelling): *central, peripheral, neurones, axons, myelin sheath, dendrites*. The aim of the activity is that the students reproduce the read text as closely as possible.

2 a dendrites b axon c neurone d nerve e nucleus f axon
g myelin sheath h nerve cell i sensory neurone j motor neurone k connector neurone

4 Voluntary responses

Voluntary responses to outside stimuli require the brain. This means that the brain is aware of what is going on around us and decides what action to take. In this type of response, the stimulus is detected by receptors in the sense organs. The receptors send a message along sensory neurones to the brain. The brain decides on a response and sends the command along motor neurones to an appropriate effector. Glands produce chemicals or muscles contract.

5 How a voluntary response works

1 a C b E c B d A e D f F g G

6 Involuntary responses

1 i, o, l, n, d, c, g, b, j, a, h, m, e, f, k, p
2 para 1: i, o, l, n, d, c, g, b (cut b after *asleep*); para 2: b, j, a, h, m, e; para 3: f, k, p
3 para 1: facts about reflexes; para 2: explanation of reflex; para 3: examples of reflexes
4 a Involuntary responses are called reflexes. Involuntary responses happen whether we want them to or not. They are a very fast kind of response, since they do not involve any decision making by the brain. We do not usually become aware of the movements that we are making in a reflex action. Reflexes can happen even while we are asleep.

Reflexes occur the following way: A stimulus is detected by receptors in a sense organ and as a result a signal is sent along sensory neurones to the spinal cord. In the spinal cord connector neurones switch in the normal response and then pass the message to the appropriate effectors. Consequently the reflex action is produced.

Examples of reflexes include the fast eye blink when a fly is just about to fly into your eye and the mouth watering when you see or smell a yummy piece of food.

b para 1: a mixture of verb types; para 2: action verbs; para 3: mixture of verbs

5 a A b F c D d B e E f C

6 F, D, B, E, C, A

7 Infectious diseases

1 smaller, rods, pain in limbs, skin rashes, rest, measles, polio, AIDS
2 They are multicellular organisms which have no roots, stems or leaves. Certain fungi produce enzymes which digest body cells. This action causes symptoms such as red, scaly, itchy patches of skin on the feet, arms, armpits or scalp. Fungal infections are usually treated with antibiotics, often in the form of a topical cream or powder. Examples of fungal infections include tinea, ringworm, thrush and Athlete's foot.



- 3 Protozoa are unicellular animals, larger than bacteria. They live in body or blood cells and cause symptoms such as diarrhoea, fever, headache and nausea. Anti-malaria drugs, such as quinine, are used to treat protozoan infections. Examples of such diseases include malaria, giardiasis, African sleeping sickness and Chagas disease.
- 4 Some infectious diseases are caused by parasitic animals, such as tapeworms, mites, fleas, ticks and liver flukes. These parasites use a host's body as a source of food, therefore irritating the body. Symptoms of infection include pain, itchiness, weight loss, diarrhoea and anaemia. Certain drugs can relieve these symptoms, and infection can be avoided by basic good hygiene and cleanliness in the home and clothing. Examples of diseases caused by parasites are Hydatid disease, scabies, flea bites and blood fluke disease.

Cause of disease	Appearance	Action	Symptoms	Treatment, prevention	Examples
a Bacteria	Microscopic Shapes like rod, ball, curve and cork screw	Toxin released that poisons and harms body	Coughing Vomiting Fever Diarrhoea Skin rashes	Hygienic handling of food and dental or surgical instruments Antibiotics	Cholera Impetigo Typhoid Whooping cough Food poisoning
b Viruses	Smaller, simpler than bacteria Rod-shaped, ball-shaped, complex geometric shapes	Take over control of cells which then can't do their job	Fever Pain in limbs Coughing Skin rashes	No treatment Rest Immunisation	Measles Flu Polio Cold sores AIDS Chicken pox

8 Non-infectious diseases

- 1 Malnutrition is caused not enough nutrients being eaten, whereas obesity is caused by too much food, and the wrong kind of food being eaten. Both diseases can also involve psychological factors as well as poor diet.

9 Hormones in the body

Name of gland	Location in body	Name of hormone	Name of target organs	Function
Pituitary	Base of brain	Human growth hormone	All organs and glands	Controls all other glands to promote growth in children.
Thyroid	Neck	Thyroxine	Body tissue	Controls metabolism.
Pancreas	Next to liver	Insulin	Blood, liver and muscle cells	Lowers the level of glucose in blood, so liver and muscles cells convert glucose to glycogen.
Adrenal	Top of kidneys	Adrenalin	Muscles, heart, liver and blood vessels	Prepares the body for action by speeding up heart beat, breathing, blood pressure and sugar levels in blood.
Ovaries (female only)	Attached to uterus	Oestrogen and progesterone	Uterus	Controls sexual function and pregnancy in female body.
Testes (male only)	Hang beneath penis outside the body	Testosterone	Penis	Controls sexual function in male body.

10 It takes two to coordinate

- 1 2 - 3 - 1 - 4

Paragraph	Main topic	Subtopics
1	Nervous and endocrine systems overview	Function
2	Nervous and endocrine systems facts	Nervous system facts Endocrine system facts
3	Examples of responses from each of the individual systems	Nerves only: Eyelid reflex, hot object reflex Hormones only: Growth, glucose uptake
4	Examples of responses from both systems together	Fright response for flight or fight

- 3 a Text is a report.
b Writing provides facts and examples about nervous and endocrine systems. The examples are explained but the overall purpose for the text is to provide information and describe the two coordination systems.



Aspect	Nervous system	Endocrine system
Purpose	To provide information to parts of the body which then respond.	
Speed	Fast	Slow
Messengers	Electrical impulse	Chemicals/ hormones
Pathway	Along neurones	Through blood vessels
Response organs	Few specific target organs	Many target organs
Response activity	Quick, short-lived	Slow, lasts for a longer time
Examples	Eye blink reflex, knee-jerk reflex, reacting to hot or sharp object	Growth

Ecology

11 Describing the environment

1 is, their, includes, carbon dioxide, temperature; Organisms, environments, live, in, where, with, their; of, biotic, an, very, has, which, in; Within, its, to, habitat, in, their, of, in; community, need, with

Term	Definition
Abiotic	Non-living features of the environment like temperature, soil and gases.
Biotic	Living aspects of the environment like animals and plants.
Community	Different groups of organisms that share the same habitat.
Ecology	Study of the living and non-living things in an environment and how they relate to each other.
Ecosystem	Living and non-living aspects depending on each other.
Environment	Surroundings.
Habitat	Place within an ecosystem where a specific living thing lives.
Population	A group of the same organism living within an ecosystem.

Object	Abiotic	Biotic
Bench, path, play equipment, buildings	✓	✗
Temperature	✓	✗
Trees, grass, water lilies	✗	✓
Pond, soil	✓	✗
Birds, duck, dog	✗	✓

- 4 a i Area 1 ii Area 3 iii Area 2
 b eg: i algae, waterlilies, frogs, little fish, mosquito wrigglers
 ii grass, clover, ants, earthworms, bees, dogs, people
 iii grass, ferns, moss, bushes, ants, birds, people

5 Answers will vary.

6 eg: Turbidity – clear, pH – 7, height of waves – 20 cm, speed of current – 5 km/h, degree of plastic pollution – high degree around the edges.

12 Look at my tree house

- 2 a gum tree b mites, insect borer, dung beetle, stick insect, possum, flying fox
 c ants, magpie, cat, tree snake, wagtail, echidna
 3 eg: gum tree → mites → ants; gum tree → stick insect → magpie; gum tree → possum → cat
 4 Insects such as ants, stick insects and mites would increase in numbers, leaving the tree infested and unhealthy.

13 The life of a nitrogen molecule

Pronoun	Noun group
you ¹	the reader
it ²	nitrogen gas
his ³	Benjamin Franklin
he ⁴	Benjamin Franklin
We ⁵	the writer & reader
his ⁶	Benjamin Franklin
It ⁷	N ₂
its ⁸	N ₂
it ⁹	NO ₂
itself ¹⁰	NO ₂
its ¹¹	NO ₂

Pronoun	Noun group
it ¹²	becoming part of a huge molecule
it ¹³	NO ₂
it ¹⁴	Protein
its ¹⁵	the bird
its ¹⁶	the bird
it ¹⁷	the bird
its ¹⁸	a bacterium
its ¹⁹	Protein
it ²⁰	NH ₃
they ²¹	the bacteria
It ²²	(no noun)

Pronoun	Noun group
its ²³	NO ₂
it ²⁴	NO ₂
It ²⁵	denitrifying bacteria
its ²⁶	N ₂
itself ²⁷	N ₂
they ²⁸	nitrifying bacteria
it ²⁹	N ₂
It ³⁰	(no noun)
He ³¹	Benjamin Franklin
our ³²	the writer & reader
It ³³	N ₂



14 Photosynthesis

- the sun, This energy, are combined, they, a green pigment, Chlorophyll, most plants, because.
- Producers make their own food; they are autotrophic.
- a carbon dioxide b chlorophyll c sun light d oxygen
- a Aim: To see whether plant leaves produce starch. Conclusion: Starch is produced in plant leaves.
b Aim: To see whether plants need light to make starch. Conclusion: Plant leaves need light to make starch.
c Aim: To see whether plant leaves need carbon dioxide to make starch. Conclusion: Plants need carbon dioxide to make starch.
d Aim: To test whether plants produce oxygen. Conclusion: Plants produce oxygen.

15 Respiration

- glucose (or sugar) + oxygen → water + carbon dioxide + energy
- a water b absorbs energy c oxygen d respiration

Question	Photosynthesis	Respiration
What does the term mean?	Producing glucose from light	Releasing energy from glucose
Which organisms perform the process?	Green plants	All living things
Is energy absorbed or released?	Absorbed	Released
What are the starting materials?	Water, carbon dioxide	Glucose, oxygen
What are the end products?	Glucose, oxygen	Water, carbon dioxide
Why is the process important?	Makes energy food, produces oxygen	Releases energy, produces water, CO ₂

- b Limewater turned milky because there was carbon dioxide in the air. The carbon dioxide came from the snails.
c The snails used up the oxygen in the air, its place was filled by limewater. The carbon dioxide produced by the snails was dissolved in the limewater.

17 Food chains and webs

Active voice	Passive voice
The rabbit eats the grass.	The grass is eaten by the rabbit.
The snake eats the mouse.	The mouse is eaten by the snake.
The small fish eat the tadpoles.	The tadpoles are eaten by the small fish.
The mouse eats the grass seeds.	The grass seeds are eaten by the mouse.
The tadpoles eat the algae.	The algae are eaten by the tadpoles.
The fox eats the rabbit.	The rabbit is eaten by the fox.
The grasshopper eats the grass seeds.	The grass seeds are eaten by the grasshopper.
The large fish eat the small fish.	The small fish are eaten by the large fish.

- a herbivore, herbivore, carnivore, carnivore, herbivore, carnivore, carnivore, herbivore, herbivore
- Healthier mango tree and fewer birds.
- a Less healthy tree and more birds.
b The birds would increase, so there would gradually be fewer insects, leading to fewer birds and a healthier tree.

18 To be adapted or not to be adapted

- a puffing up its body
b lays large number of eggs, lays eggs in fresh or salt water, eats insects and canned food
c rough, brown, dull coloured skin, skin with poison glands

Situation	Adaptation	Category	Purpose
Snow	Heavy set, roundish body	Physical	Stay warm, not lose much heat
	Dressed in warm clothes	Behavioural	Keep warm, not lose body heat
	Shivers	Physiological	Produce heat, become warm
Beach	Sweats	Physiological	Cool down, lose body heat
	Wears swimmers	Behavioural	Lose heat easily
	Wears hat	Behavioural	Not get hot
Central Australia	Dark skin	Physical or Physiological	Not get sun burnt easily
	Skinny, tall	Physical	Lose heat easily
	Rest in middle of day	Behavioural	Not get too hot



3

Long sentence	Type of adaptation
a Elephants have large ears, as a result they can cool off quickly.	physical
b When kangaroos lick their paws, their spit evaporates and so the paws become cooler.	behavioural
c Lizards move to the shade on hot days so they stay cool.	behavioural
d Eagles have sharp, curved beaks, therefore they can tear meat.	physical
e Because rabbits rest in burrows, they stay cool.	behavioural
f Since dolphins have a streamlined body, they can swim fast.	physical
g Birds have light bones and wings and therefore they can fly.	physical
h Because kookaburras have a tough beak, they can eat snakes.	physical
i The cactus has spines for leaves and as a result water is conserved, as they have no pores.	physical
j Dogs shiver on cold mornings, as a result they get warm.	physiological
k Since koalas sleep in the middle of the day, they don't get too hot.	behavioural
l Frogs have slimy moist skin so they can breathe through their skin.	physical

19 Design an animal

- It is difficult to observe marsupial moles, because they spend most of their time underground.
- Would eat worms, larvae underground and insects above ground; find food by smelling them.
- Scientists may have kept a marsupial mole in an enclosure with dry sand and a water dish. They noticed that the water level did not go down.

20 Case study – the imaginary bush

- Case 1:** The wallabies would feed on the small bush which would become weaker and smaller, perhaps even die. The possums would fertilise more flowers increasing seed/berry formation. Foxes would eat some wallabies and possums relieving the pressure on the small bush.
- Case 2:** The birds from another valley would eat the berries and carry the seeds to their valley where the seeds may grow into more small bushes.
- Case 3:** Feral dogs would hunt the wallabies for prey. The small bush can grow bigger because its young leaves are not eaten by the wallabies.
- Case 4:** Many birds and possums are displaced and have to find somewhere else to live. As a result the berries are not distributed which keeps the new small trees in the valley. The flowers are not fertilised and berries are not produced. Consequently few new small trees will develop.
- Case 5:** The shy small bird does not eat the berries and so the seeds are not spread far. As a result new small bushes grow near the older ones.
- Case 6:** The small bushes would grow better because there are no wallabies to eat their young leaves. The birds would not spread the seeds far and new bushes would grow near the old ones. Because the campers take away the flowers, possums lose that source of food and would go elsewhere to live. The bushes are damaged by the campers which could lead to the bush dying. Soon the small bush and much animal wildlife would have disappeared.
- Case 7:** The eagles would catch fewer possums and birds. As a result more flowers of the small bush would get fertilised and the seeds would be spread over a larger area resulting in more bushes.
- Case 8:** Many small bushes would be damaged and possibly destroyed resulting in fewer bushes.
- Case 9:** The small bushes would grow healthier and in greater numbers and the wildlife would return.

21 Down the slime and up the ladders

- eg Burning fossil fuels to make electricity.
- eg Deforestation (ie clearing trees) means that there are fewer trees to produce carbon dioxide.
- & d Teacher to check.

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